Advanced Empirical Finance: Topics and Data Science

Volatility estimation

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Why do we care about volatility?

- We can decompose any time series into a predictable part and an unpredictable part

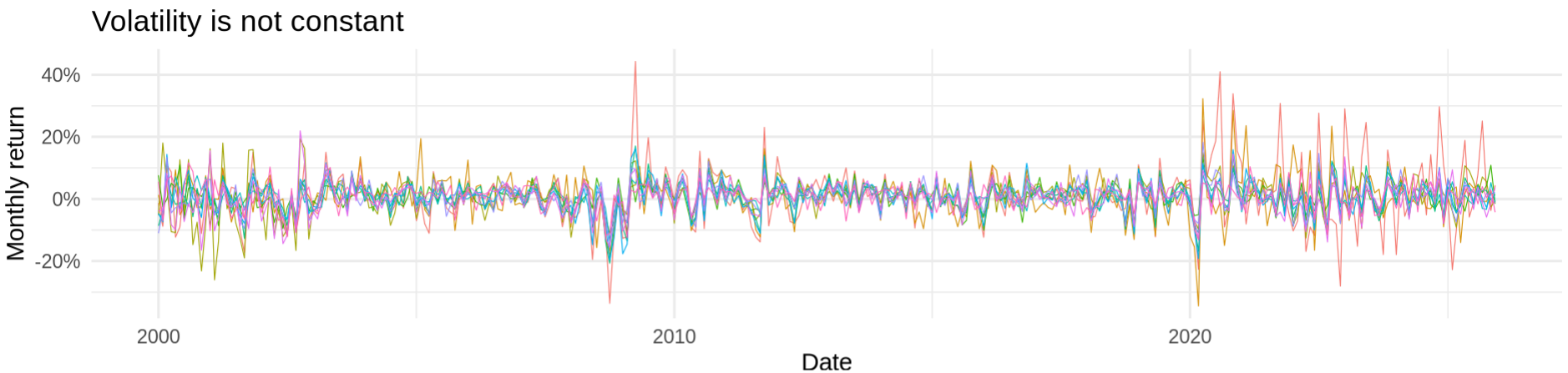
$$r_t = E\left(r_t|\mathcal{F}_{t-1}\right) + \varepsilon_t$$

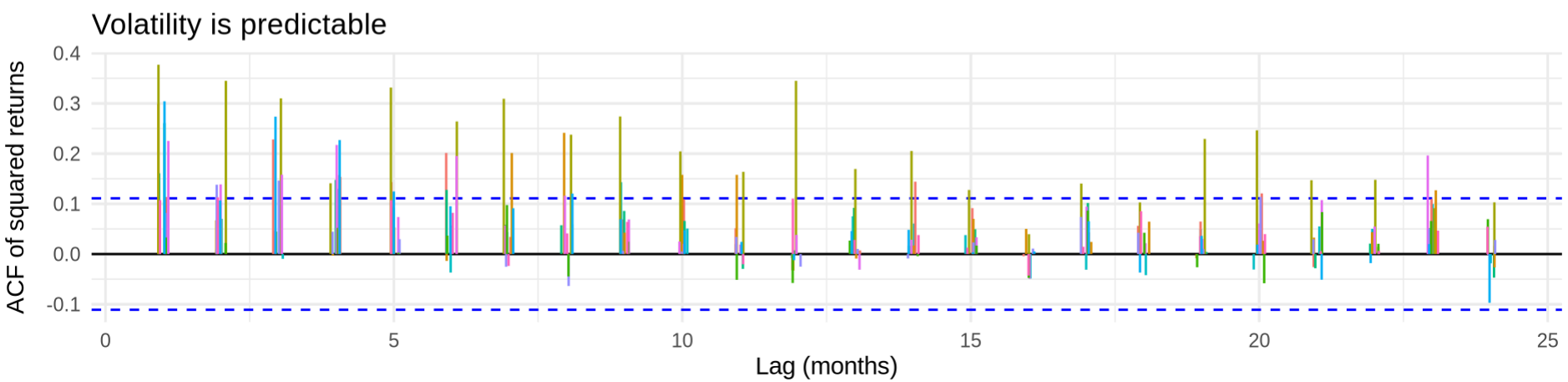
  where $\mathcal{F}_{t-1}$ denotes all information available at the end of day $t-1$
- In the last part of the lecture we focused on the conditional mean $\mu_t = E\left(r_t|\mathcal{F}_{t-1}\right)$
- This part: The conditional variance of r_t

$$\sigma_t^2 = \text{Var}\left(r_t|\mathcal{F}_{t-1}\right) = E\left(\varepsilon_t^2|\mathcal{F}_{t-1}\right)$$

- $\hat{\sigma}_t^2$ is a crucial component for *option pricing* - recall Black-Scholes
- $\hat{\Sigma}_t$ is relevant for the *portfolio choice problem*
- The core issue: σ_t^2 is not observable — at each date we only see one realization r_t, not the distribution that generated it

What do we know about σ_t^2?





Naive estimators

Simplest idea: a **rolling-window** sample variance over the last M observations, where M is the window length (e.g. $M = 60$ for a 3-month daily window) and $\bar{r}_{t,M}$ is the sample mean over the same window:

$$\hat{\sigma}_t^2 = \frac{1}{M-1} \sum_{s=t-M+1}^{t} (r_s - \bar{r}_{t,M})^2$$

- The **cliff-edge problem:** when a large return exits the window, $\hat{\sigma}_t^2$ jumps even though no new information arrived today
- The bias-variance tradeoff: short M reacts fast but is noisy; long M is smooth but stale

The **RiskMetrics** estimator (J.P. Morgan 1994) with smoothing parameter $\lambda \in (0, 1)$ (RiskMetrics default: $\lambda = 0.94$):

$$\hat{\sigma}_t^2 = (1-\lambda)\, r_{t-1}^2 + \lambda\, \hat{\sigma}_{t-1}^2$$

- No cliff-edge: RiskMetrics includes all past returns with exponentially declining weights

Naive estimators



Conditional Heteroscedastic Models

- The manner under which σ_t^2 evolves over time distinguishes one volatility model from another
- I will focus on conditional heteroscedastic models that use an exact function to govern the evolution of σ_t^2, but there are also models with stochastic components (stochastic volatility)

1. ARCH (autoregressive conditional heteroscedasticity)
2. GARCH (generalized ARCH)

Volatility model building[1]

Building a volatility model for an asset return series consists of four steps:

1. Specify a mean equation (e.g., multi-factor model) or test for serial dependence in the data and, if necessary, build an econometric model (e.g., an ARMA model) for the return series to remove any linear dependence
2. Use the residuals of the mean equation to test for autoregressive conditional heteroskedasticity (ARCH) effects
3. Specify a volatility model if ARCH effects are statistically significant, and perform a joint estimation of the mean and volatility equations
4. Check the fitted model carefully and refine it if necessary

The ARCH model (Engle, 1982)

- An ARCH(m) model describes the dependence of ε_t by

$$\varepsilon_t = \sigma_t z_t \qquad \sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + ... + \alpha_m \varepsilon_{t-m}^2$$

- where z_t is a sequence of independent and identically distributed (iid) random variables with mean zero and unit variance
- (common assumption for MLE: Gaussian or t-distributed z_t)
- Intuition: large past shocks ε_{t-1}^2 imply higher conditional variance (clustering)

Parameter considerations:

- The sequence of parameters α_i must satisfy some conditions to guarantee positivity of σ_t^2: $\alpha_0 > 0$, and $\alpha_i \geq 0 \forall i > 0$
- The coefficients α_i must satisfy some regularity conditions to ensure that the unconditional variance of ε_t is finite / σ_t^2 is stationary (all roots in k of $1 - \alpha_1 k - ... - \alpha_m k^m = 0$ lie outside the unit circle)
- The unconditional variance equals $\sigma_\varepsilon^2 = \frac{\alpha_0}{1 - \sum_{i=1}^m \alpha_i}$

The GARCH model (Bollerslev, 1986)

- Problem of the ARCH model: typical financial time series often require highly parameterized ARCH models
- We define the GARCH(p,q) model as

$$\varepsilon_t = \sigma_t z_t \qquad \sigma_t^2 = \alpha_0 + \sum_{j=1}^{p} \alpha_j \varepsilon_{t-j}^2 + \sum_{\mathbf{i=1}}^{\mathbf{q}} \beta_{\mathbf{i}} \sigma_{\mathbf{t-i}}^{\mathbf{2}}$$

- We can write any stationary GARCH(p,q) model as an ARCH(∞) model

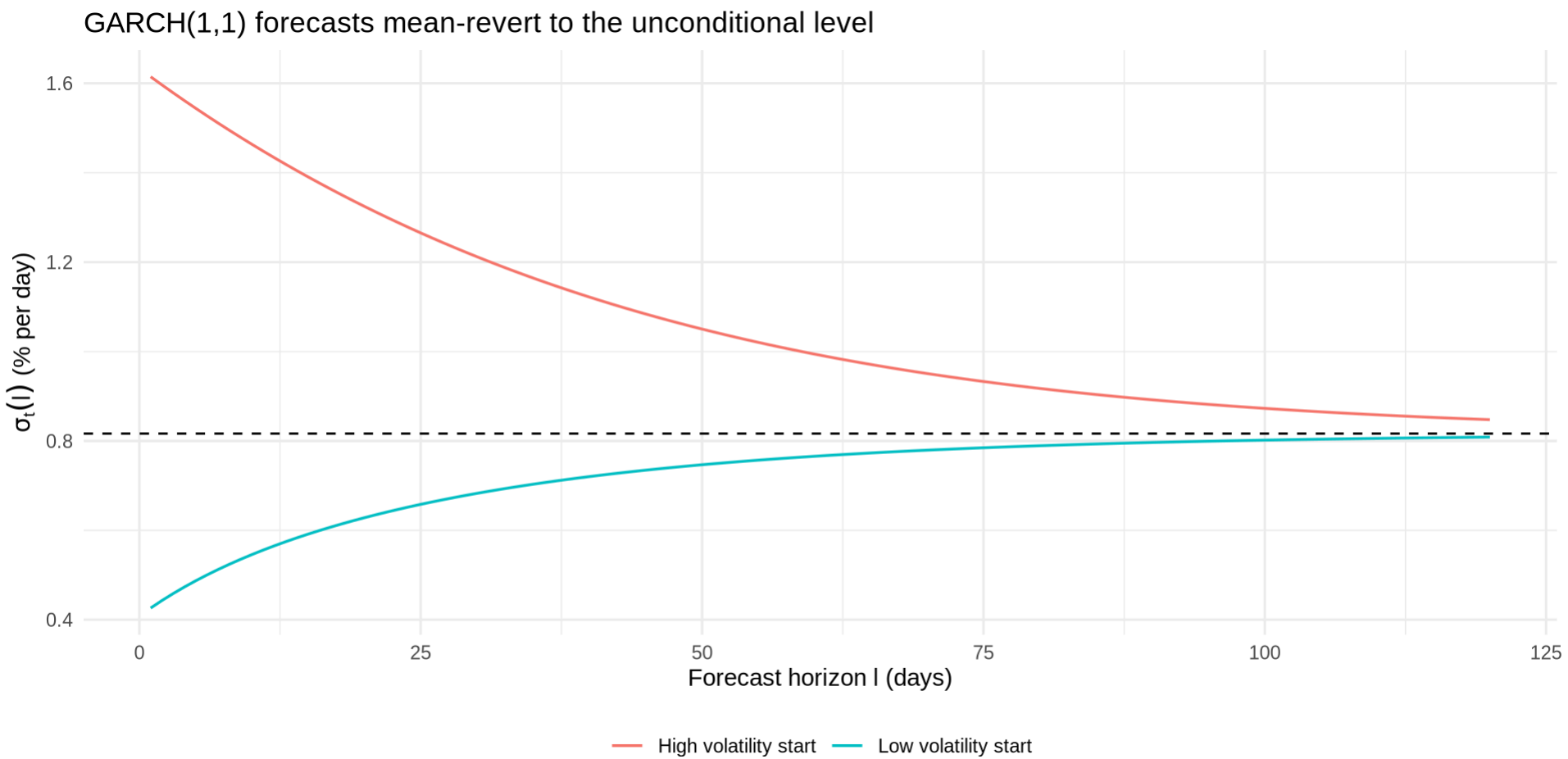
Parameter considerations:

- The conditions $\alpha_0 \geq 0, \alpha_j \geq 0, \beta_i \geq 0$ ensure positive σ_t^2
- Similar conditions for stationarity as for ARCH: all roots in k of $1 - \alpha_1 k - \ldots - \alpha_p k^p - \beta_1 k - \ldots - \beta_q k^q = 0$ lie outside the unit circle

Forecasting

Forecasting: illustration

- **Intuition:** volatility forecasts decay geometrically toward the unconditional variance $\bar{\sigma}^2 = \alpha_0/(1 - \alpha_1 - \beta_1)$ at rate $\alpha_1 + \beta_1$ – today's volatility shock matters most for the near future and vanishes as the horizon grows



Model estimation

- Log likelihood function under the normality assumption for ε_t

$$\begin{aligned}\log \mathcal{L}(\theta) &= \log f(r_1|\theta) + \log f(r_2|r_1,\theta) + \ldots + \log f(r_T|r_{t-1},\ldots,r_1,\theta) \\ &= -\frac{T}{2}\log(2\pi) - \sum_{t=1}^{T}\left(\log(\sigma_t^2) + \frac{(r_t-\mu_t)^2}{2\sigma_t^2}\right)\end{aligned}$$

  where we compute σ_t^2 and μ_t recursively

- Similar of course for different likelihood specifications, e.g. Student t (consult Tsay (2010) for specific estimation issues)
- Consult the exercises for hands-on examples

1. Define (negative) log-likelihood function
2. Use `optim` (R) or `minimize` (Python) to find maximum likelihood estimator (in some cases, you need constrained optimization)
3. Retrieve information matrix (`hessian = TRUE` (R) or `hess_inv` (Python))
4. Compute standard errors

```r
#R
fit <- optim(..., hessian = TRUE) # minimizes the negative log-likelihood
# The inverse Hessian estimates the covariance matrix of the estimator
standard_errors <- sqrt(diag(solve(fit$hessian)))
```

Fitting GARCH in Practice

`tsgarch` (Galanos) is the modern R package for GARCH estimation.

The key choice is the **distribution** of the innovations z_t: a skewed Student-t captures fat tails and asymmetry, while a plain Gaussian likelihood yields consistent parameter estimates but requires robust standard errors for inference.

R | Python

```r
manuf_returns <- returns |>
  filter(industry == "manuf")
returns_xts <- xts(manuf_returns$return, order.by = manuf_returns$date)

spec <- garch_modelspec(
  y            = returns_xts,
  model        = "garch",
  order        = c(1, 1),
  distribution = "sstd"
)

fit <- estimate(spec)
summary(fit)
```

Value-at-risk: definition

- We define the VaR (Value-at-risk) of a financial position over a given time horizon l with tail probability p by

$$p = \Pr\left(L(l) \geq VaR\right) = 1 - \Pr\left(L(l) \leq VaR\right)$$

  where L is a loss function, e.g. in terms of USD

- Intuition: the probability that the position holder would encounter a loss greater than or equal to VaR over the time horizon l is at most p



Small example: Value-at-risk



Multivariate volatility estimation

Multivariate volatility estimation

- Covariances are relevant for portfolio theory (and therefore asset pricing, capital budgeting, investment decisions)
- Requires estimation of Σ_t instead of just σ_t^2

Problems for estimates $\hat{\Sigma}_t$:

1. The curse of dimensionality (too many parameters)
2. Time variation (volatilities and correlations may change over time)
3. Frequency (minute, daily, monthly data?)

Discuss: How would you compute minimum variance portfolio weights for a universe of 1000 assets?

General Framework

- We consider the N-dimensional return series

$$r_t = E\left(r_t | \mathcal{F}_{t-1}\right) + \varepsilon_t$$

  where ε_t is a $(N \times 1)$ vector of error terms with (conditional) covariance matrix Σ_t

- Exploiting the symmetry property of Σ_t we can write

$$\Sigma_t = D_t Q_t D_t$$

  where Q_t is the $(N \times N)$ conditional correlation matrix of ε_t and $D_t = \mathrm{diag}\left(\sigma_{11,t}, \ldots, \sigma_{NN,t}\right)$ is a $(N \times N)$ diagonal matrix

- Because of the symmetry of Σ_t, it is sufficient to consider only $N(N+1)/2$ unique elements

The curse of dimensionality

- Consider the case with a $(T \times N)$ matrix of returns R where $T < N$
- Then the sample variance covariance matrix is

$$S = \frac{1}{T-1} R' \left(I_T - \frac{1}{T} \iota \iota' \right) R$$

- The rank of the matrix $R'R$ is $\mathbf{min}(T, N) = T$
- (intuition: $R'R$ qua linear transformation sends $\mathbb{R}^N$ into a subspace with dimension at most T)
- This implies that $R'R$ is singular for $N \geq T$ and thus

$$\exists w \in \mathbb{R}^N : w' S w = 0$$

- In other words, S is not invertible and there is no unique efficient portfolio
- We need more structure to overcome this issue (e.g. based on factor structure)

Linear shrinkage (Ledoit Wolf, 2003, 2004)

- The sample covariance matrix is unbiased and the maximum likelihood estimator under normality
- In the context of mean estimation, Stein (1956) showed that we can construct a better estimator than the sample mean by shrinking the sample mean to a target vector, that is, by using a linear combination of the sample mean and the target vector
- Better in the sense of the mean squared error (MSE) — a little bias in exchange for a large reduction in variance
- Ledoit and Wolf (2003, 2004) apply the same thinking to Σ
- They suggest a linear combination of some target matrix F (identity, single factor, or equicorrelation matrix) and the sample variance-covariance matrix $\hat{\Sigma}$, where the *shrinkage intensity* α controls the weights:

$$\hat{\Sigma}^{LW} = \alpha F + (1 - \alpha)\hat{\Sigma} \text{ for } 0 \leq \alpha \leq 1$$

The shrinkage target (Ledoit Wolf, 2004)

- We already discussed (multi) factor models
- Constant correlation model: we assume that all the (pairwise) correlations are identical.
- Estimator of the common constant correlation: average of all the sample correlations

Illustrating shrinkage

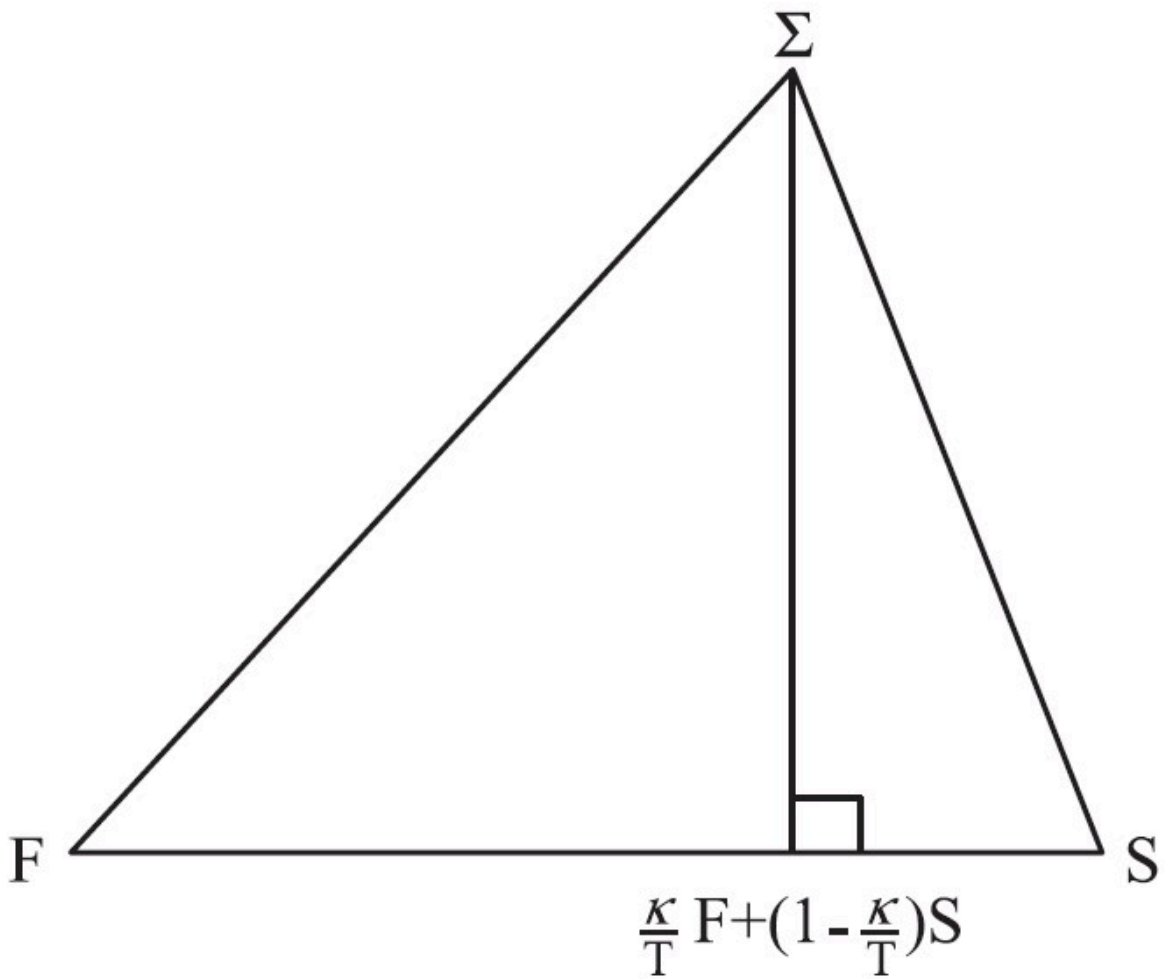


Fig. 1. Geometric interpretation of Theorem 1. The notion of orthogonality among *N*-dimensional symmetric matrices is defined by the inner product associated with the Frobenius norm.

Figure 1

- The optimal linear combination depends on unknown population quantities and, therefore, we must think of it as an ideal (or “oracle”) but infeasible estimator
- Ledoit and Wolf derive feasible estimators (an exercise asks you to implement the estimator)

The optimal shrinkage intensity (Ledoit-Wolf 2003, 2004)

Dynamic multivariate models

- The generalization of univariate GARCH models to the multivariate domain is conceptually simple
- We can broadly divide the literature on the dynamics governing Σ_t into direct multivariate extensions, factor models, linear combinations of univariate GARCH models (Generalized Orthogonal GARCH), and nonlinear combinations of univariate GARCH models (the broader class of Dynamic Correlation models)
- The need to invert the covariance matrix in many MGARCH parameterizations introduces estimation problems for large systems, as the eigenvalues of the covariance matrix decrease exponentially fast toward zero, even when the covariance is not singular

Constant conditional correlation (CCC)

- Bollerslev (1990)
- In the constant conditional correlation (CCC) model of Bollerslev (1990), we can decompose the covariance matrix into

$$\Sigma_t = D_t R D_t = \rho_{ij} \sqrt{h_{iit} h_{jjt}}$$

  where $D_t = \mathrm{diag}(\sqrt{h_{11t}}, \dots, \sqrt{h_{NNt}})$ and R is a positive definite correlation matrix
- Maximum likelihood estimation possible in the multivariate normal case (see Eq. 50 in Boudt et al., 2019)

Obvious extension: DCC

- Covariance Targeting (Engle (2002))

$$\Sigma_t = D_t \mathbf{R_t} D_t = \rho_{ij} \sqrt{h_{iit} h_{jjt}}$$

- DCC is a **two-step estimator**: (i) fit a univariate GARCH to each asset to obtain D_t and the standardized residuals $z_t = D_t^{-1} \varepsilon_t$; (ii) model the dynamics of the correlation matrix using a proxy process Q_t:

$$Q_t = \bar{Q} + \alpha \left(z_t z_t' - \bar{Q} \right) + \beta \left(Q_{t-1} - \bar{Q} \right)$$

  where $\bar{Q}$ is the sample covariance of the z_t and (α, β) are scalar parameters governing the correlation dynamics

- We then obtain the correlation matrix R_t as

$$R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}$$

Large-scale portfolio allocation under model uncertainty and transaction costs

Bridging to large-scale allocation

- At time t, we solve the problem

$$\max_{\omega_{t+1}} \ \omega_{t+1}'\mu_t - \frac{\gamma}{2}\omega_{t+1}'\,\Sigma_t\,\omega_{t+1} \quad s.t. \quad \omega_{t+1}'\,\iota = 1,$$

  where Σ_t denotes the $N \times N$ asset return covariance, μ_t the mean vector, γ the relative risk aversion, and ι a vector of ones

- Optimal portfolio weights:

$$\omega_{t+1}^* = \frac{1}{\iota'\Sigma_t^{-1}\iota}\Sigma_t^{-1}\iota + \frac{1}{\gamma}\left(\Sigma_t^{-1} - \frac{1}{\iota'\Sigma_t^{-1}\iota}\Sigma_t^{-1}\iota\iota'\Sigma_t^{-1}\right)\mu_t$$

... the classical problem

- Noisy estimates of μ_t over short horizons
- Solution (?): Fixing μ_t (e.g. $\mu_t = 0$) or using long horizons
- $(N \times N)$ matrix Σ_t contains $N(N+1)/2$ distinct elements $\Rightarrow$ Estimation error
- For $N = 500$: 125,250 elements
- Sample covariance only positive definite if $T \geq N$

$\Rightarrow$ Plugging in sample estimates $\hat{\mu}_t$ and $\hat{\Sigma}_t$ performs badly!

Solution: Regularizing Σ ...

- Shrinking the covariance matrix (Ledoit and Wolf, 2004):

$$\Sigma_t = \hat{\alpha} F_t + (1 - \hat{\alpha}) S_t,$$

where S_t is the sample variance-covariance matrix and F_t denotes the equi-correlation matrix

Alternative approaches: - Restricting portfolio weights (*Jagannathan and Ma, 2003*; Fan et al, 2012) - Factor models (MacKinlay Pastor, 2000; Fan et al, 2008) - Parsimonious high-dimensional parametric volatility models (Engle Ledoit Wolf, 2017) - Imposing priors (Barry, 1974; Brown, 1979; Jorion, 1985, 1986; Black Litterman, 1992; Pastor Stambaugh, 2000)

... or using high-frequency data:

- Idea: Constructing a daily covariance estimate using HF data of the same day
- E.g., Liu (2009), Hautsch et al (2012), Hautsch et al (2015), Lunde et al (2015), Bollerslev et al (2016), ...

But ...

Frictions:

- No consideration of transaction costs!
- Re-balancing is costly!

Estimation:

- Parameters are uncertain!
- Model ambiguity
- Predictive ability of models is time-varying!
- De Miguel et al (2009): Out of sample, no strategy can outperform the $1/N$ portfolio in high dimensions!

Portfolio optimization under transaction costs

The investor at time t observes past returns $R_t = (r_1', \dots, r_t')' \in \mathbb{R}^{t \times N}$ and assumes an underlying (predictive) model $\mathcal{M}$.

The (myopic) portfolio optimization problem is

$$\begin{aligned}\omega_{t+1}^* :=& \arg \max_{\omega \in \mathbb{R}^N, \iota'\omega = 1} E\left(U_\gamma\left(\omega'\left(1 + r_{t+1}\right) - \nu_t(\omega)\right) | \mathcal{M}, \mathcal{F}_t\right) \\ =& \arg \max_{\omega \in \mathbb{R}^N, \iota'\omega = 1} \int U_\gamma\left(\omega'\left(1 + r_{t+1}\right) - \nu_t(\omega)\right) p_t(r_{t+1}|\mathcal{M}) \, \mathrm{d}r_{t+1},\end{aligned}$$

where $p_t(r_{t+1}|\mathcal{M}) := p_t(r_{t+1}|\mathcal{M}, \mathcal{F}_t)$ denotes the predictive return distribution and $\nu_t(\omega)$ the transaction costs.

A closed-form solution for Gaussian returns and quadratic transaction costs

- Initial Markowitz (1952) approach with Gaussian returns: $p_t(r_{t+1}|\mathcal{M}) = N(\mu, \Sigma)$
- We can solve it by maximizing the certainty equivalent

$$\omega_{t+1}^* = \arg \max_{\omega \in \mathbb{R}^N, \iota'\omega = 1} \omega'\mu - \nu_t(\omega) - \frac{\gamma}{2}\omega'\Sigma\omega.$$

- We assume **quadratic** transaction costs — mainly because they yield a closed-form solution that is directly comparable to the classical Markowitz problem (Hautsch and Voigt (2019) show that we can treat the L_1 case similarly):

$$\nu(\omega_{t+1}, \omega_{t^+}, \beta) = \frac{\beta}{2}(\omega_{t+1} - \omega_{t^+})'(\omega_{t+1} - \omega_{t^+}),$$

  with cost parameter $\beta > 0$
- The pre-rebalancing weights ω_{t^+} are *not* the same as yesterday's target weights ω_t: even if the investor does nothing, prices move during the day and drift the portfolio away from ω_t. Concretely:

$$\omega_{t^+} := \frac{\omega_t \circ (1 + r_t)}{\iota'(\omega_t \circ (1 + r_t))}$$

Optimization problem[1]

- We leave the proof as an exercise

$$\begin{aligned}\omega_{t+1}^{*} :=& \arg \max_{\omega \in \mathbb{R}^N, \iota'\omega = 1} \omega'\mu - \nu_t(\omega, \omega_{t^+}, \beta) - \frac{\gamma}{2}\omega'\Sigma\omega \\ =& \arg \max_{\omega \in \mathbb{R}^N,\ \iota'\omega = 1} \omega'\mu^{*} - \frac{\gamma}{2}\omega'\Sigma^{*}\omega,\end{aligned}$$

where the effective mean and covariance absorb the transaction-cost terms:

$$\mu^{*} := \mu + \beta\omega_{t^+} \quad \text{and} \quad \Sigma^{*} := \Sigma + \frac{\beta}{\gamma} I_N$$

- As a result:

$$\omega_{t+1}^{*} = \frac{1}{\gamma}\left(\Sigma^{*-1} - \frac{1}{\iota'\Sigma^{*-1}\iota}\Sigma^{*-1}\iota\iota'\Sigma^{*-1}\right)\mu^{*} + \frac{1}{\iota'\Sigma^{*-1}\iota}\Sigma^{*-1}\iota$$

What happens if $\beta \to \infty$?

- Define $A(\Sigma) := \left(\Sigma^{-1} - \frac{1}{\iota'\Sigma^{-1}\iota}\Sigma^{-1}\iota\iota'\Sigma^{-1}\right)$.
- Proposition. Let ω_0 be the initial allocation. Then,

$$\omega_T^* = \sum_{i=0}^{T-1}\left(\frac{\beta}{\gamma}A(\Sigma^*)\right)^i \omega(\mu, \Sigma^*) + \left(\frac{\beta}{\gamma}A(\Sigma^*)\right)^T \omega_0,$$

  where $\omega(\mu, \Sigma^*)$ denotes the mean-variance efficient allocation.
- Proposition.

$$\lim_{\beta\to\infty} \omega_{t+1}^* = \left(I_N - \frac{1}{N}\iota\iota'\right)\omega_{t^+} + \frac{1}{N}\iota = \omega_{t^+}$$

Efficiency of the portfolio

- Proposition. $\exists \beta^* > 0 \ \forall \tilde{\beta} \in [0, \beta^*) : \left\| \left(\frac{\tilde{\beta}}{\gamma} A(\Sigma^*) \right) \right\|_F < 1$, where $\| \cdot \|_F$ denotes the Frobenius norm $\|A\|_F := \sqrt{\sum_{i=1}^{N} \sum_{j=1}^{N} a_{i,j}^2}$
- For $\beta < \beta^*$ and $T \to \infty$, we have

$$\sum_{i=0}^{T} \left(\frac{\beta}{\gamma} A(\Sigma^*) \right)^i \to \left(I_N - \frac{\beta}{\gamma} A(\Sigma^*) \right)^{-1}$$

  and $\lim_{i \to \infty} \left(\frac{\beta}{\gamma} A(\Sigma^*) \right)^i = 0$.
- Proposition. For $T \to \infty$ and $\beta < \beta^*$ the series ω_T converges to a unique fixed point

$$\omega_\infty = \left(I_N - \frac{\beta}{\gamma} A(\Sigma^*) \right)^{-1} \omega(\mu, \Sigma^*) = \omega(\mu, \Sigma)$$

Intuition behind shrinkage

- For asset-specific transaction costs

$$(\omega_{t+1} - \omega_{t^+})' \, B \, (\omega_{t+1} - \omega_{t^+})$$

  where B is positive definite, we can reconcile the linear shrinkage estimator for Σ of Ledoit and Wolf (2003, 2004).

- Lemma. Assume a regime of high volatility with $\Sigma^h = (1+h)\Sigma$, where $h > 0$ and Σ is the asset return covariance matrix during calm periods. Assume $\mu = 0$. Then, the optimal weight ω^*_{t+1} is equivalent to the optimal portfolio based on Σ and (reduced) transaction costs $\frac{\beta}{1+h} < \beta$.

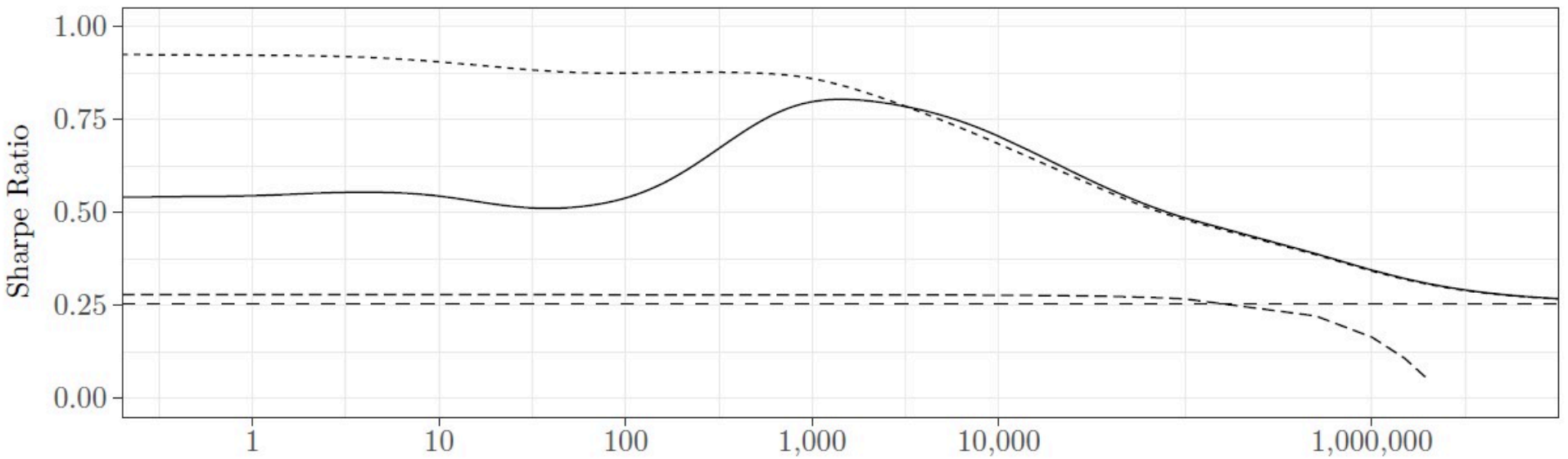
Empirical Illustration

- $N = 308$ assets; 06/2007 - 03/2017 with daily readjustments
- We estimate Σ_t with
    1. Sample covariance
    2. Ledoit/Wolf (2004) shrinkage estimator
    3. Rolling window of length $h = 500$ days
- We fix μ_t to zero. Initial portfolio weights: $\frac{1}{N}\iota$. $\gamma = 4$.
- Yields realized portfolio returns net of transaction costs

$$r_t^\beta - \beta \left(\omega_{t+1}^\beta - \omega_{t^+}^\beta\right)' \left(\omega_{t+1}^\beta - \omega_{t^+}^\beta\right)$$

with $r_t^\beta := r_t' \omega_t^\beta$.

Realized Sharpe ratios with quadratic transaction costs



- Sample (solid line), LW (dotted line), Equal (dashed)
- X-axis denotes transaction cost parameter β in basis points

L_1 Transaction Costs

- Usual assumption in the literature: L_1 transaction costs

$$\nu_t\left(\omega, \omega_{t^+}, \beta\right) = \beta||\omega - \omega_{t^+}||_1 = \beta \sum_{i=1}^{N} |\omega_{i,t+1} - \omega_{i,t^+}|.$$

- For $\mu = 0$ and $\Delta := \omega_{t+1} - \omega_{t^+}$, we have

$$\omega_{t+1}^* = \arg \min_{\Delta \in \mathbb{R}^N,\ \iota'\Delta = 0} \gamma \Delta' \Sigma \omega_{t^+} + \frac{\gamma}{2} \Delta' \Sigma \Delta + \beta ||\Delta||_1,$$

- Solving for ω_{t+1}^* yields

$$\omega_{t+1}^* = \left(1 + \frac{\beta}{\gamma} \iota' \Sigma^{-1} \tilde{g}\right) \omega_{\text{mvp}} - \frac{\beta}{\gamma} \Sigma^{-1} \tilde{g},$$

where $\tilde{g}$ is the vector of sub-derivatives of $||\omega_{t+1} - \omega_{t^+}||_1$ and $\omega_{\text{mvp}} := \frac{1}{\iota' \Sigma^{-1} \iota} \Sigma^{-1} \iota$ are the weights of the GMV portfolio.

Recall Jagannathan and Ma?

- Proposition. For $\mu = 0$, the underlying problem is equivalent to

$$\omega_{t+1}^* = \arg \min_{\omega \in \mathbb{R}^N,\ \iota'\omega = 1} \omega' \Sigma_{\frac{\beta}{\gamma}} \omega,$$

where $\Sigma_{\frac{\beta}{\gamma}} = \Sigma + \frac{\beta}{\gamma}\left(g^* \iota' + \iota g^{*'}\right)$ and g^* is the subgradient of $||\omega_{t+1}^* - \omega_{t^+}||_1$.

What about parameter uncertainty?

- Recall that we compute the optimal portfolio (for now: $\nu = 0$) as the solution to

$$\begin{aligned}\omega_{t+1}^{*} :=& \arg\max_{\iota'\omega=1} E\left(U_{\gamma}\left(\omega'\left(1+r_{t+1}\right)\right)|\mathcal{M},\mathcal{F}_{t}\right)\\ =& \arg\max_{\iota'\omega=1} \int U_{\gamma}\left(\omega'\left(1+r_{t+1}\right)\right) p_{t}(r_{t+1}|\mathcal{M})\,\mathrm{d}r_{t+1}\\ =& \arg\max_{\iota'\omega=1} \int\int_{\mu}\int_{\Sigma} U_{\gamma}\left(\omega'\left(1+r_{t+1}\right)\right) p_{t}(r_{t+1}|\mu,\Sigma,\mathcal{M})\,\mathrm{d}\mu\,\mathrm{d}\Sigma\,\mathrm{d}r_{t+1},\end{aligned}$$

  where $p_t(r_{t+1}|\mathcal{M}) := p_t(r_{t+1}|\mathcal{M},\mathcal{F}_t)$ denotes the predictive return distribution

- Unlike the conditional distribution, the Bayesian predictive distribution accounts for estimation errors by integrating over the unknown parameter space
- The problem above refers to **Bayesian Portfolio Allocation**

1. Account for estimation risk and model uncertainty
2. Elicit economically meaningful prior beliefs about asset pricing models (distribution of future returns)
3. Highly flexible due to Bayesian computation techniques

Bayesian asset allocation[1]

- So far we analysed a two-step procedure

1. Estimate the parameters $\hat{\theta}$ (e.g. $\hat{\mu}_t$ and $\hat{\Sigma}_t$)
2. Maximize the expected utility conditional on the estimated parameters being the true ones

$$\max_{\omega} = E\left(U(\omega)|\theta = \hat{\theta}\right)$$

- Optimally, however, the investor adjusts for estimation uncertainty
- Thus, portfolio selection based on $p_t(r_{t+1}|\mathcal{M})$ instead of $p_t(r_{t+1}|\hat{\theta}, \mathcal{M})$

Example: Bayesian Portfolio Allocation

- Specify the prior distribution of the parameters as diffuse

$$\pi(\theta) = p(\mu_t|\Sigma_t) \times \pi(\Sigma_t) \propto |\Sigma_t|^{-\frac{N+1}{2}}$$

- Then, assuming that the returns are jointly normally distributed, we obtain the posterior distribution

$$p(\mu_t|\Sigma_t, \mathcal{D}_t) \propto |\Sigma_t|^{-1/2} \exp\left(-\frac{1}{2} tr\left(T(\mu_t - \hat{\mu}_t)\right)(\mu_t - \hat{\mu}_t)'\Sigma_t^{-1}\right)$$

and

$$\pi(\Sigma_t) \propto |\Sigma_t|^{-(T+N)/2} \exp\left(-\frac{1}{2} tr\Sigma_t^{-1}\left(T\hat{\Sigma}_t\right)\right)$$

- The predictive distribution corresponds to a multivariate t distribution with $T - N$ degrees of freedom
- Under the diffuse prior the mean-variance efficient portfolio is

$$\omega^{\text{Bayes}} = \frac{1}{\gamma}\left(\frac{T - N - 2}{T + 1}\right)\hat{\Sigma}_t^{-1}\hat{\mu}_t$$

- Intuition: Invest less in the risky asset and more in the risk-free asset

Extensions and takeaways

- Bayesian computation allows us to sample from (almost) arbitrary posterior predictive distributions
- We can derive the optimal allocation numerically
- Turnover regularization is closely related to parameter shrinkage
- Transaction costs shift optimal portfolios towards current holdings
- This effect is different from *statistical* approaches: Shrinkage intensity is an exogenous parameter!

Application: Hautsch and Voigt (2019)

	$\hat{\mu}$	$\hat{\sigma}$	SR	CE	TO	pc	sp	% Trade
$\omega^{\text{SV, no TC}}$	-58.9 [-62.7,-55.8]	16.12 [15.72,16.52]	-	-	55.73 [53.52,58.16]	31.102 [30.279,31.942]	2.41 [2.12,2.78]	> 99.95
$\omega^{\text{HF, no TC}}$	< -99	15.97 [15.52,16.33]	-	-	245.4 [239.29,250.43]	56.791 [55.7,57.892]	4.01 [3.83,4.65]	> 99.95
ω^{Sample}	5.4 [3.8,7.2]	16.50 [15.87,17.16]	0.326 [0.224,0.446]	3.993 [2.367,5.908]	0.19 [0.18,0.20]	42.427 [37.75,47.465]	1.21 [0.98,1.38]	18.45 [18.20,19.74]
ω^{LW}	5.6 [4.5,6.6]	14.72 [14.28,15.06]	0.380 [0.307,0.449]	4.501 [3.412,5.497]	0.88 [0.83,0.93]	24.405 [22.784,26.427]	0.98 [0.65,1.11]	11.95 [10.92,12.96]
ω^{HF}	6.5 [5.9,6.9]	16.71 [16.52,16.93]	0.388 [0.354,0.414]	5.081 [4.488,5.518]	0.14 [0.09,0.21]	17.165 [15.937,18.062]	0.48 [0.21,0.61]	14.44 [13.41,15.54]
ω^{SV}	6.6 [5.9,7.0]	16.76 [16.59,16.99]	0.391 [0.354,0.417]	5.153 [4.489,5.589]	0.32 [0.21,0.41]	17.860 [16.086,18.827]	0.71 [0.68,0.92]	14.13 [13.11,15.20]
$\omega^{\text{Comb.}}$	6.5 [5.0,7.8]	14.86 [14.45,15.39]	0.438 [0.342,0.52]	5.401 [3.877,6.693]	0.49 [0.46,0.53]	20.609 [18.483,23.115]	0.81 [0.54,0.92]	13.70 [12.53,14.83]
ω^{Naive}	5.2 [4.9,5.5]	23.5 [23.14,23.84]	0.224 [0.204,0.234]	2.399 [2.035,2.733]	1.10 [1.09,1.12]	0.4 [0.4,0.4]	-	> 99.95
$\omega^{\text{Naive, 2m}}$	5.9 [5.7,6.2]	23.03 [22.68,23.36]	0.258 [0.245,0.273]	3.294 [2.991,3.632]	0.15 [0.14,0.16]	0.403 [0.403,0.404]	-	1.66 [1.63,1.66]
ω^{mvp}	-59.4 [-62.8,-56.1]	14.11 [13.73,14.55]	-	-	53.49 [52.25,54.64]	45.65 [42.864,48.28]	3.33 [3.24, 3.42]	> 99.95
$\omega^{\text{mvp, no s.}}$	2.8 [2.0,3.6]	13.1 [12.97, 13.29]	0.213 [0.149, 0.279]	1.935 [1.092, 2.794]	3.53 [3.34, 3.69]	10.616 [9.161, 11.548]	-	> 99.95
$\omega^{\text{kz, 3f}}$	< -99	> 50	-	-	941.9 [175.51,2339.7]	> 1000	5.98 [3.72, 8.8]	> 99.95
ω^{tz}	< -99	> 50	-	-	> 1000	> 1000	28.09 [17.31, 57.51]	> 99.95
ω^{bs}	< -99	> 50	-	-	> 1000	> 1000	124.7 [39.8, 228.14]	> 99.95
ω^{ϑ}	-24.28 [-26.28, -22.15]	12.04 [11.78, 12.34]	-	-	25.846 [25.43, 26.31]	18.692 [18.05, 19.28]	1.499 [1.40, 1.58]	> 99.95
ω^{Market}	6.2	21.19	0.292	3.958	-	-	-	-

Table 2: Annualized averages of the bootstrapped out-of-sample portfolio performances based on 1904 trading days. Transaction costs are proportional to the L_1 norm of re-balancing (as of (12)). $\hat{\mu}$ is the annualized portfolio return in percent, $\hat{\sigma}$ is the annualized standard deviation in percent, SR denotes the (annualized) out-of-sample Sharpe ratio of the individual strategies, CE is the Certainty Equivalent for an investor with power utility function and risk-aversion factor $\gamma = 4$, TO is the average turnover in percent, pc is the average weight concentration (L_2 norm of the portfolio weights) and sp is the average proportion of the sum of

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Parametric Portfolio Choice

Stock characteristics and optimal portfolio choice

- Idea: parametrize weights as a function of the characteristics such that we maximize expected utility
- feasible for large portfolio dimensions (such as the entire CRSP universe)
- Brandt et al. (2009) proposed it in their influential paper *Parametric Portfolio Policies: Exploiting Characteristics in the Cross Section of Equity Returns*.

Basic idea:

- At each date t we have N_t stocks in the investment universe, where each stock i has return $r_{i,t+1}$ and a vector of firm characteristics $x_{i,t}$ such as time-series momentum or market capitalization
- The investor's problem is to choose portfolio weights $w_{i,t}$ to maximize the expected utility of the portfolio return

$$\max_{w} E_t\left(U(r_{p,t+1})\right) = E_t\left[U\left(\sum_{i=1}^{N_t} w_{i,t} r_{i,t+1}\right)\right]$$

Where do the stock characteristics show up?

- Parametrize the optimal portfolio weights as a function of $x_{i,t}$ with the following linear specification for the portfolio weights:

$$w_{i,t} = \bar{w}_{i,t} + \frac{1}{N_t}\theta' \hat{x}_{i,t}$$

- here, $\bar{w}_{i,t}$ is the weight of a benchmark portfolio (value-weighted or naive portfolio), θ is a vector of coefficients which we are going to estimate, and $\hat{x}_{i,t}$ are the characteristics of stock i, which we standardize cross-sectionally to have zero mean and unit standard deviation. **Note on notation:** the hat on $\hat{x}_{i,t}$ denotes *standardization*, not an estimator — an unusual convention we borrow from Brandt et al. (2009)
- Think of the portfolio strategy as a form of active portfolio management relative to a performance benchmark: The individual stock characteristics determine the deviations from the benchmark portfolio
- By construction the weights sum up to one as $\sum_{i=1}^{N_t} \hat{x}_{i,t} = 0$ due to the standardization. Note that the coefficients are constant across assets and through time
- Implicit assumption is that the characteristics fully capture all aspects of the joint distribution of returns that are relevant for forming optimal portfolios

Optimal Choice of θ

- We estimate θ by replacing the expectation with its sample analogue and *maximizing the sample-average realized utility* of the implied portfolio returns:

$$\hat{\theta} = \arg\max_{\theta} \frac{1}{T} \sum_{t=0}^{T-1} U\left(\sum_{i=1}^{N_t} \left(\bar{w}_{i,t} + \frac{1}{N_t} \theta' \hat{x}_{i,t} \right) r_{i,t+1} \right)$$

- In practice this is a low-dimensional numerical optimization (e.g. via `optim` in R) — the dimension of θ equals the number of characteristics, independent of N_t
- The allocation strategy is attractive because the number of parameters to estimate is small
- Instead of a tedious specification of the N_t-dimensional vector of expected returns and the $N_t(N_t + 1)/2$ free elements of the variance-covariance matrix, all we need to focus on in our application is the vector θ

What about short-selling constraints?

- Enforcing portfolio constraints via the parametrization is not straightforward, but in our case we simply renormalize the portfolio weights before returning them by computing

$$w_{i,t}^{+} = \frac{\max(0, w_{i,t})}{\sum_{j=1}^{N_t} \max(0, w_{j,t})}$$